

BUSFIN 8200 - Problem Set 1

Maeve Nguyen

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Problem 1

1 (a)

Starting from $R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}$ and assuming that D is greater than zero, we take log of both sides and have:

$$\begin{aligned} r_{e,t+1} &= \log(P_{t+1} + D_{t+1}) - \log(P_t) \\ &= \log(P_{t+1}) \left(1 + \frac{D_{t+1}}{P_{t+1}}\right) - p_t \\ &= p_{t+1} + \log\left(1 + \frac{e^{d_{t+1}}}{e^{p_{t+1}}}\right) - p_t \\ &= p_{t+1} + \log(1 + e^{d_{t+1} - p_{t+1}}) - p_t \\ &= p_{t+1} + \log(1 + e^{dp_{t+1}}) - p_t \\ &= p_{t+1} - p_t + f(dp_{t+1}) \end{aligned}$$

where $f(dp_{t+1})$ is a log function of the log dividend-price ratio. Applying a first-order Taylor expansion around $\bar{dp}$ to $f(dp_{t+1})$, we have Equation (1):

$$r_{e,t+1} \approx p_{t+1} - p_t + f(\bar{dp}) + f'(\bar{dp})(dp_{t+1} - \bar{dp})$$

We define $\kappa = \frac{1}{1 + e^{\bar{dp}}}$. Consequently, we have the following:

$$\begin{aligned} f'(\bar{dp}) &= \frac{e^{\bar{dp}}}{1 + e^{\bar{dp}}} = 1 - \kappa \\ e^{\bar{dp}} &= \frac{1}{\kappa} - 1 \\ \bar{dp} &= \log\left(\frac{1}{\kappa} - 1\right) \end{aligned}$$

Plugging in the above results into Equation (1), we have:

$$\begin{aligned} r_{e,t+1} &\approx p_{t+1} - p_t + \log(1 + e^{\bar{dp}}) + (1 - \kappa)dp_{t+1} - (1 - \kappa)\log\left(\frac{1}{\kappa} - 1\right) \\ &= p_{t+1} - p_t + \log\left(\frac{1}{\kappa}\right) - (1 - \kappa)p_{t+1} + (1 - \kappa)d_{t+1} - (1 - \kappa)\log\left(\frac{1}{\kappa} - 1\right) \\ &= \kappa p_{t+1} - p_t + (1 - \kappa)d_{t+1} + \kappa_0 \\ &= \kappa_0 - \kappa dp_{t+1} + (d_{t+1} - d_t) + (d_t - p_t) \\ &= \kappa_0 - \kappa dp_{t+1} + \Delta d_{t+1} + dp_t \end{aligned}$$

where $\kappa_0 = \log\left(\frac{1}{\kappa}\right) - (1 - \kappa)\log\left(\frac{1}{\kappa} - 1\right) = -\log(\kappa) - (1 - \kappa)\log\left(\frac{1}{\kappa} - 1\right)$. Rearranging the above equation, we have:

$$\begin{aligned} dp_t &= -\kappa_0 + \kappa dp_{t+1} + r_{e,t+1} - \Delta d_{t+1} \\ \Rightarrow dp_{t+1} &= -\kappa_0 + \kappa dp_{t+2} + r_{e,t+2} - \Delta d_{t+2} \\ \Rightarrow dp_{t+2} &= -\kappa_0 + \kappa dp_{t+3} + r_{e,t+3} - \Delta d_{t+3} \end{aligned}$$

Keeping substituting these equations into the first one and generalizing the pattern H times, we have:

$$\begin{aligned}
dp_t &= -\kappa_0 \underbrace{(1 + \kappa + \kappa^2)}_{\sum_{h=1}^H \kappa^{h-1}} + \kappa^3 dp_{t+3} + [r_{e,t+1} + \kappa r_{e,t+2} + \kappa^2 r_{e,t+3}] - [\Delta d_{t+1} + \kappa \Delta d_{t+2} + \kappa^2 \Delta d_{t+3}] \\
&= -\kappa_0 \left(\frac{1 - \kappa^H}{1 - \kappa} \right) + \kappa^H \cdot dp_{t+H} + \sum_{h=1}^H \kappa^{h-1} \cdot r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \cdot \Delta d_{t+h} \\
&= \kappa_0 \left(\frac{\kappa^H - 1}{1 - \kappa} \right) + \kappa^H \cdot dp_{t+H} + \sum_{h=1}^H \kappa^{h-1} \cdot r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \cdot \Delta d_{t+h} \tag{1.1}
\end{aligned}$$

Given that $\mathbb{E}_t[dp_t] = dp_t$, we take conditional expectation on both sides of Equation (1.1) and have:

$$dp_t = \kappa_0 \left(\frac{\kappa^H - 1}{1 - \kappa} \right) + \kappa^H \cdot \mathbb{E}_t[dp_{t+H}] + \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] \tag{1.2}$$

Since $e^{\bar{dp}}$ is always positive, we have $0 < \kappa < 1$. Therefore, as $H \rightarrow \infty$, $\kappa^H \rightarrow 0$. In addition, we also need to impose No-Bubble Condition such that $\lim_{H \rightarrow \infty} \kappa^H \cdot \mathbb{E}_t[dp_{t+H}] = 0$. Giglio, Maggiori, and Stroebel (2016) indeed shows that there were no systemic deviations in the discounts of extremely long leaseholds to otherwise identical freeholds in the U.K. and Singapore housing markets, thus indicating no rational bubbles. Consequently, letting $H \rightarrow \infty$, Equation (1.2) can be simplified as follows:

$$dp_t = \frac{-\kappa_0}{1 - \kappa} + \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] \tag{1.3}$$

1 (b)

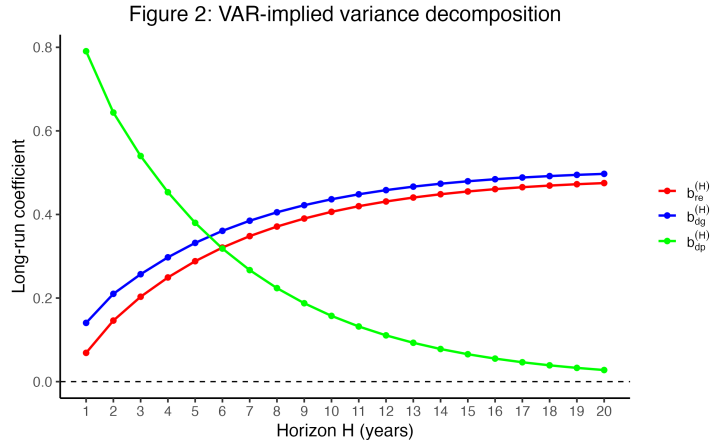
Using Equation 1.4 from the problem set, I decomposed the variation in the log dividend-price ratio into variations associated with the discounted future returns, discounted future dividend growth and terminal dividend-price ratio. By constructing these regressors at a 12-month increment, I regressed each component on dp_t in univariate OLS models to estimate their respective betas through different horizons as shown below.



The figure indicates that over the short horizon (up to 5 years), the discounted future dividend-price ratio is persistent and responsible for the majority of the variation in the log dividend-price ratio, as demonstrated by a strongly positive $\beta_{dp}^{(H)}$. Indeed, at $H = 1$, this component explains 80% of the variation in dp_t . However, as the horizon extends beyond year 1, $\beta_{dp}^{(H)}$ gradually declines and approaches zero at $H = 12$, before turning negative for the rest of the horizons. This further proves the no-bubble condition imposed in part (a). Meanwhile, from $H = 4$ onward, the discounted future return component quickly rises and becomes the dominant force in explaining the variability in dp_t , with $\beta_{re}^{(H)}$ reaching 0.9 at $H = 20$. This further supports the notion that the discount rate component is more relevant in explaining long-run, rather than short-run, variation in the log dividend-price ratio. As for the dividend growth component, $\beta_{dg}^{(H)}$ remains relatively flat, hovering around 0.2 before gradually converging to 0.1 at $H = 20$. While the log dividend-price ratio has some relationship with discounted future dividend growth, it is considerably weaker than its relationship with discounted future expected returns, especially in the long horizons. Indeed, this evidence is consistent with Cochrane's address to Stambaugh, stating that log dividend-price ratio's inability to forecast future dividend growth proves that stock returns are predictable.

1 (c)

Applying Equation 1.4 to a VAR(1) model, I constructed Figure 2 as below.



In particular, let $z_t = [dg_t, r_{e,t}, dp_t]$ follow $z_{t+1} = \Gamma_0 + \Gamma z_t + \varepsilon_{t+1}$ where $\varepsilon_{t+1} \stackrel{\text{i.i.d.}}{\sim} (0, \Sigma)$. Running these overlapping annual regressions, I recover $\hat{\Gamma}$ in the form of $(z_t' z_t)^{-1} z_t' z_{t+1}$ to forecast three state variables in the future. Then, I regress dg_t , $r_{e,t}$, and dp_t onto dp_t using univariate full-sample OLS specifications to recover their coefficients in the form of the b_z vector. As a result, I calculate the VAR-implied coefficients as follows:

$$\begin{aligned}
b_{dg}^{(H)} &= -\mathbf{1}_{dg}'(\Gamma - \kappa^H \cdot \Gamma^{H+1})(I - \kappa \cdot \Gamma)^{-1} b_z, \\
b_{re}^{(H)} &= \mathbf{1}_{re}'(\Gamma - \kappa^H \cdot \Gamma^{H+1})(I - \kappa \cdot \Gamma)^{-1} b_z, \\
b_{dp}^{(H)} &= 1 - b_{dg}^{(H)} - b_{re}^{(H)}
\end{aligned}$$

1 (d)

Using the same VAR process in equation (1.5) and taking H to ∞ , the term $\kappa^H \cdot dp_{t+H}$ should go to 0 under the no-bubble condition. However, there is still a residual VAR-implied b_{dp}^∞ , albeit minor at 0.0011919. This could be attributed to the first-order specifications of the VAR process. Otherwise, the VAR-implied covariance coefficients for the discount rate and dividend growth components are as follows:

Table 1: VAR-implied infinite-horizon coefficients

Component	Coefficient
$b_{re}^{(\infty)}$	0.489191
$b_{dg}^{(\infty)}$	0.509617

Table 1 suggests that under the VAR(1) specifications, eventually both expected return and dividend growth equally contribute to the variation in the log dividend-price ratio in the extremely long run.

1 (e)

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}.$$

Taking log of both sides, we have:

$$\begin{aligned}
 r_{e,t+1} &= \log(P_{t+1} + D_{t+1}) - p_t \\
 &= \log\left(P_{t+1} + \frac{P_{t+1} \cdot D_{t+1}}{P_{t+1}}\right) - p_t \\
 &= \log\left(P_{t+1} \left(1 + \frac{D_{t+1}}{P_{t+1}}\right)\right) - p_t \\
 &= p_{t+1} + \underbrace{\log\left(1 + \frac{D_{t+1}}{P_{t+1}}\right)}_{\equiv dy_{t+1}} - p_t
 \end{aligned} \tag{1.4}$$

We also have:

$$\begin{aligned}
 dy_t &= \log\left(1 + \frac{D_t}{P_t}\right) = \log\left(1 + \frac{e^{d_t}}{e^{p_t}}\right) = \log(1 + e^{dp_t}) \\
 \Rightarrow -dy_t &= -\log(1 + e^{dp_t}) = \log(1 + e^{dp_t})^{-1} \\
 \Rightarrow e^{-dy_t} &= e^{\log(1 + e^{dp_t})^{-1}} = \frac{1}{1 + e^{dp_t}} \\
 \Rightarrow \frac{1}{e^{-dy_t}} - 1 &= e^{dp_t} \\
 \Rightarrow \frac{1 - e^{-dy_t}}{e^{-dy_t}} &= e^{dp_t} \\
 \Rightarrow 1 - e^{-dy_t} &= e^{dp_t} \cdot e^{-dy_t} \\
 \Rightarrow \log(1 - e^{-dy_t}) &= dp_t - dy_t
 \end{aligned} \tag{1.5}$$

Rewriting Equation (1.4) and plugging in Equation (1.5), we have:

$$\begin{aligned}
 r_{e,t+1} - (d_{t+1} - d_t) &= (d_t - p_t) - (d_{t+1} - p_{t+1}) + dy_{t+1} \\
 \Rightarrow r_{e,t+1} - \Delta d_{t+1} &= dp_t - dp_{t+1} + dy_{t+1} \\
 &= (dp_t - dy_t) - (dp_{t+1} - dy_{t+1}) + dy_t \\
 &= \log(1 - e^{-dy_t}) - \log(1 - e^{-dy_{t+1}}) + dy_t
 \end{aligned} \tag{1.6}$$

Applying a first-order Taylor approximation to $\log(1 - e^{-dy_t})$ around $\overline{dy}$, we have:

$$\log(1 - e^{-dy_t}) = f(dy_t) \approx f(\overline{dy}) + f'(\overline{dy})(dy_t - \overline{dy}).$$

Define $\kappa \equiv e^{-\bar{dy}}$. We have the following:

$$\begin{aligned} f(\bar{dy}) &= \log(1 - e^{-\bar{dy}}) = \log(1 - \kappa), \\ f'(\bar{dy}) &= \frac{1}{1 - e^{-\bar{dy}}} \cdot e^{-\bar{dy}} = \frac{e^{-\bar{dy}}}{1 - e^{-\bar{dy}}} = \frac{\kappa}{1 - \kappa}. \end{aligned}$$

As a result,

$$f(dy_t) = \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \bar{dy}).$$

Similarly,

$$f(dy_{t+1}) = \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_{t+1} - \bar{dy}).$$

Plugging these two equations into Equation (1.6), we have:

$$\begin{aligned} \Rightarrow \quad r_{e,t+1} - \Delta d_{t+1} &= \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \bar{dy}) - \log(1 - \kappa) - \frac{\kappa}{1 - \kappa}(dy_{t+1} - \bar{dy}) + dy_t \\ &= \frac{\kappa}{1 - \kappa}(dy_t - dy_{t+1}) + dy_t \\ &= \frac{\kappa + 1 - \kappa}{1 - \kappa}dy_t - \frac{\kappa}{1 - \kappa}dy_{t+1} \\ &= \frac{1}{1 - \kappa}dy_t - \frac{\kappa}{1 - \kappa}dy_{t+1} \\ &= \frac{dy_t - \kappa dy_{t+1}}{1 - \kappa}. \end{aligned}$$

$$\Rightarrow \quad dy_t = (1 - \kappa)(r_{e,t+1} - \Delta d_{t+1}) + \kappa dy_{t+1}.$$

Recursively substituting dy_{t+h} , we have:

$$dy_t = (1 - \kappa) \cdot \left[\sum_{h=1}^H \kappa^{h-1} \cdot r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \cdot \Delta d_{t+h} \right] + \kappa^H \cdot dy_{t+H} \quad (1.7)$$

Taking the conditional expectation of both sides and assuming that $\mathbb{E}_t[X] = X$, we have:

$$dy_t = (1 - \kappa) \cdot \left[\sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] \right] + \kappa^H \cdot dy_{t+H} \quad (1.8)$$

Knowing that $0 < \kappa < 1$ by construction and assuming that the no-bubble condition holds such that $\lim_{H \rightarrow \infty} \kappa^H \cdot dy_{t+H} = 0$, we let $H \rightarrow \infty$ in Equation (1.8) and have:

$$dy_t = (1 - \kappa) \left[\sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] \right] \quad (1.9)$$